

Clinical Study Supporting Document

Drug Information

Drug Name: Evimoxin (adalimumab biosimilar) **Manufacturer:** BioPharm Innovations **Reference Product:** Humira (adalimumab)

Study Details

Study ID: EVX-RA-2023-01 **Study Type:** Phase III Clinical Trial **Study Date:** January 2023 - December 2023

Medical Condition

Target Condition: Rheumatoid Arthritis (Moderate to Severe) **Patient Population:** Adult patients (18-75 years) with moderate to severe rheumatoid arthritis with inadequate response to methotrexate therapy

Study Design

Design Type: Randomized, double-blind, active-controlled study **Number of Participants:** 352 patients (175 in Evimoxin group, 177 in reference group) **Treatment Duration:** 24 weeks **Primary Endpoint:** ACR20 response at Week 24

Secondary Endpoints:

- ACR50 and ACR70 responses
- DAS28-CRP change from baseline
- Safety and immunogenicity assessment

Dosage Information

Evimoxin Dosage: 40 mg subcutaneous injection every other week **Reference Product Dosage:** 40 mg subcutaneous injection every other week **Concomitant Treatments:** Stable dose of methotrexate (10-25 mg per week)

Key Results

Primary Endpoint Results:

- Evimoxin group: 72.6% achieved ACR20 at Week 24

- Reference product group: 71.2% achieved ACR20 at Week 24

Secondary Endpoint Results:

- ACR50 at Week 24: 45.7% (Evimoxin) vs. 44.1% (reference)
- ACR70 at Week 24: 22.3% (Evimoxin) vs. 21.5% (reference)
- Mean change in DAS28-CRP: -2.1 (Evimoxin) vs. -2.0 (reference)

Safety Profile

Treatment-Emergent Adverse Events (TEAEs):

- Evimoxin group: 56.0% of patients reported at least one TEAE
- Reference product group: 58.2% of patients reported at least one TEAE

Most Common TEAEs (>5% in either group):

1. Injection site reactions (4.6% Evimoxin, 5.1% reference)
2. Upper respiratory tract infections (8.0% Evimoxin, 7.9% reference)
3. Headache (5.7% Evimoxin, 6.2% reference)
4. Nausea (3.4% Evimoxin, 4.0% reference)

Serious Adverse Events:

- Evimoxin group: 3.4% (6 patients)
- Reference product group: 3.4% (6 patients)

Discontinuations Due to Adverse Events:

- Evimoxin group: 2.3% (4 patients)
- Reference product group: 2.8% (5 patients)

Immunogenicity

Anti-Drug Antibody (ADA) Development:

- Evimoxin group: 11.4% (20 patients)
- Reference product group: 12.4% (22 patients)

Neutralizing Antibody Development:

- Evimoxin group: 5.7% (10 patients)
- Reference product group: 6.2% (11 patients)

Pharmacokinetics

Key PK Parameters (Week 12):

- C_{max}: 4.9 µg/mL (Evimoxin) vs. 5.1 µg/mL (reference)
- AUC_{0-336h}: 1,820 µg·h/mL (Evimoxin) vs. 1,870 µg·h/mL (reference)
- t_{1/2}: 14.2 days (Evimoxin) vs. 14.0 days (reference)

Conclusions

The results of this study demonstrate that Evimoxin is highly similar to its reference product Humira in terms of efficacy, safety, immunogenicity, and pharmacokinetic properties in patients with moderate to severe rheumatoid arthritis. The observed differences between the biosimilar and reference product were not clinically meaningful, supporting the conclusion that Evimoxin is a biosimilar to Humira.

The benefit-risk profile of Evimoxin is favorable and comparable to that of the reference product. This study supports the use of Evimoxin as a biosimilar alternative to Humira for the treatment of rheumatoid arthritis in adult patients.

References

1. Anderson J, et al. Treatment of rheumatoid arthritis with biologics: 2020 update. *Rheumatology*. 2020;59(1):55-70.
2. Chen B, et al. Comparative effectiveness of TNF inhibitors in rheumatoid arthritis. *Autoimmun Rev*. 2022;21(6):103071.
3. FDA Guidance for Industry: Scientific Considerations in Demonstrating Biosimilarity to a Reference Product (2015).
4. EMA Guideline on similar biological medicinal products containing monoclonal antibodies (2012).

Document Metadata

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